

Grade 12 - Functions and Inverses

27 January 2026

Memorandum

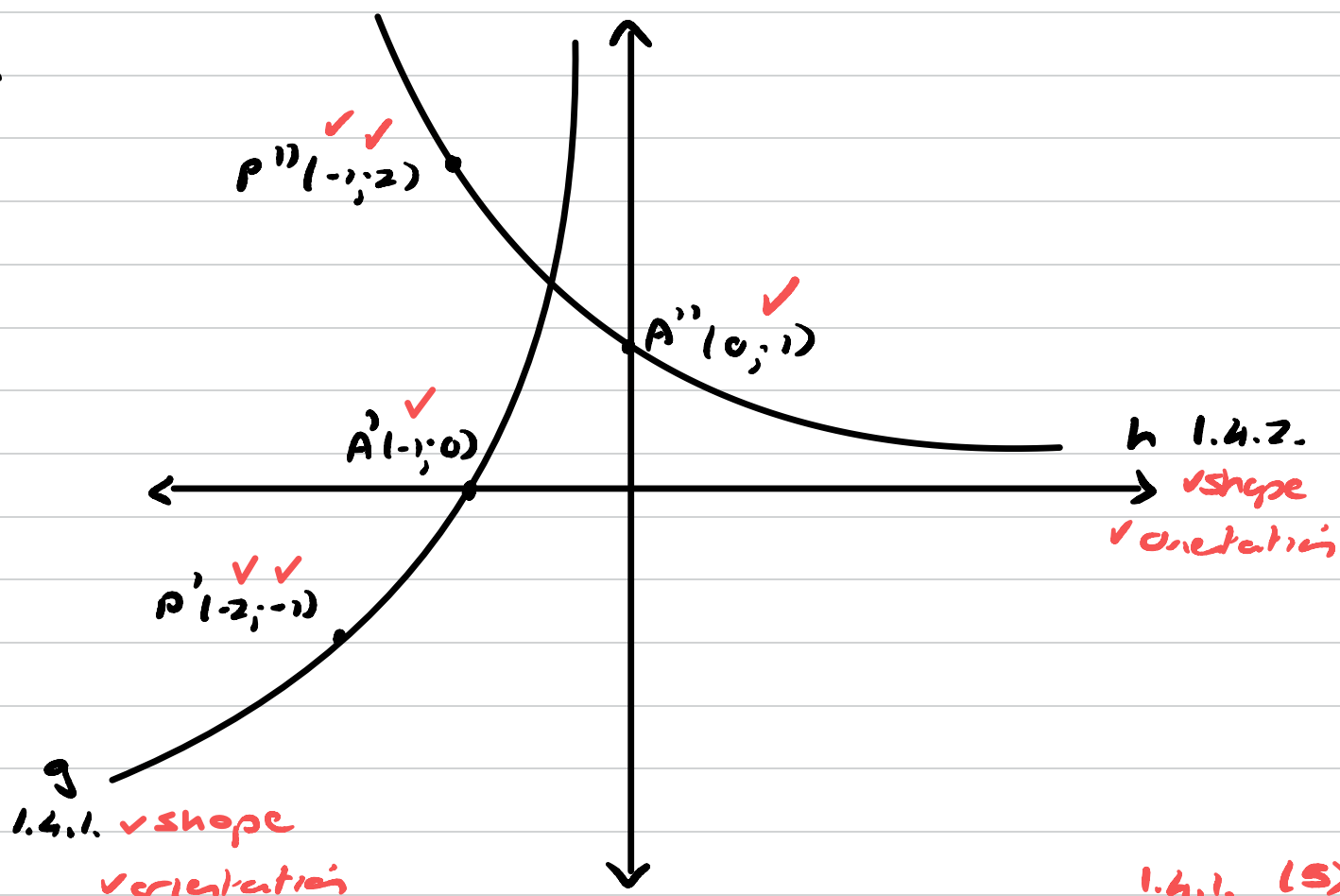
Task 1

1.1. $x > 0$ ✓ A (1)

1.2. $A(1; 0)$ ✓ x -value ✓ A only if in coordinate form (2)

1.3. $y > 0$ ✓ A (1)

1.4.



1.4.1. (5)

1.4.2. (5)

1.5. $x < -1$ ✓ A ✓ A (2)

Task 2

2.1. $y = x$

2.2. $y = -\sqrt{-\frac{1}{3}x} ; x \leq 0 ; y \geq 0$

Let $y = x$ ✓ M

$$\therefore x = -\sqrt{-\frac{1}{3}y} ; y \leq 0 ; x \leq 0$$

$$\therefore -x = \sqrt{-\frac{1}{3}y} ; y \leq 0 ; x \leq 0$$

$$\therefore x^2 = -\frac{1}{3}y ; \text{✓ A } y \leq 0 ; x \leq 0$$

$$\therefore -3x^2 = y ; y \leq 0 ; x \leq 0$$

$$\therefore f'(x) = -3x^2 ; x \leq 0 \quad \text{✓ CA only if correct restriction included}$$

(3)

2.3. $-3x^2 = x$ ✓ M
 $0 = 3x^2 + x$
 $0 = x(3x+1)$
 $x = 0$ or $x = -\frac{1}{3}$
✓ A ✓ CA if same as x

$$\therefore A(-\frac{1}{3}, -\frac{1}{3})$$

(3)

$$2.4. \quad y = -\frac{1}{3}x^2 ; x \leq 0$$

$$\text{Let } x = -x \quad \checkmark 14$$

$$y = -\frac{1}{3}(-x)^2 ; -x \leq 0$$

$$y = -\frac{1}{3}x^2 ; x \geq 0$$

$$g(x) = -\frac{1}{3}x^2 ; x \geq 0 \quad \checkmark CA \quad \checkmark A \quad (13)$$

$$2.5. \quad y = -6x + 3c$$

$$\text{sub } (1; 2c) \quad \checkmark 14$$

$$2c = -6(1) + 3c$$

$$6 = c \quad \checkmark A$$

$\therefore$ Point of intersection is $(1; 12)$ ✓ CA

$$h(x) = -3(x-1)^2 + q \quad \checkmark 14 \text{ either}$$

$\therefore$ Turningpoint is $(1; q)$

$$\therefore q = 12 \quad \checkmark CA$$

(15)

Task 3

A
✓ gradient

3.1.

$$y = x + c$$

sub (1, -2)

✓ M substitution

$$-2 = 1 + c$$

$$c = -3$$

$$\therefore C (0, -3)$$

(2)

3.2.

$$y = \frac{a}{x-1} - 2$$

✓ sub value of p and q

sub (0, -3) ✓ M substitute

$$\therefore -3 = \frac{a}{0-1} - 2$$

$$\therefore -1 = -a$$

✓ A value of a

$$\therefore a = 1$$

(3)

3.3.

$$y = x - 3$$

$$\text{Let } y = 0$$

$$0 = x - 3$$

$$3 = x \quad \checkmark A$$

$$\therefore B(3, 0)$$

$$y = \frac{1}{x-1} - 2$$

$$\text{Let } y = 0$$

$$0 = \frac{1}{x-1} - 2$$

$$2 = \frac{1}{x-1}$$

$$2x - 2 = 1$$

$$2x = 3$$

$$\therefore x = \frac{3}{2} \quad \checkmark A$$

$$\therefore A\left(\frac{3}{2}, 0\right)$$

$$AB = 3 - \frac{3}{2} \quad \checkmark CA$$

$$= \frac{3}{2} \text{ units}$$

(4)

3.4.

$$\frac{1}{x-1} - 2 = x - 3 \quad \checkmark M$$

$$\frac{1}{x-1} = x - 1$$

$$1 = (x-1)^2 \quad \checkmark A$$

$$\therefore x-1 = 1 \quad \text{or} \quad x-1 = -1$$

$$x = 2$$

$$x = 0$$

$$y = 2 - 3 \quad \checkmark M$$

$$y = -1$$

$$E(2, -1) \quad \checkmark CA \quad \checkmark CA \text{ means only if in coordinate form.}$$

(5)

3.5.1

$\checkmark A$ $\checkmark CA$ $\checkmark$ value
 $1 < x \leq \frac{3}{2}$

(2)

3.5.2.

$\checkmark A$
 $x < 0$ or

$\checkmark A$ $\checkmark CA$
 $1 < x \leq 2$

(3)